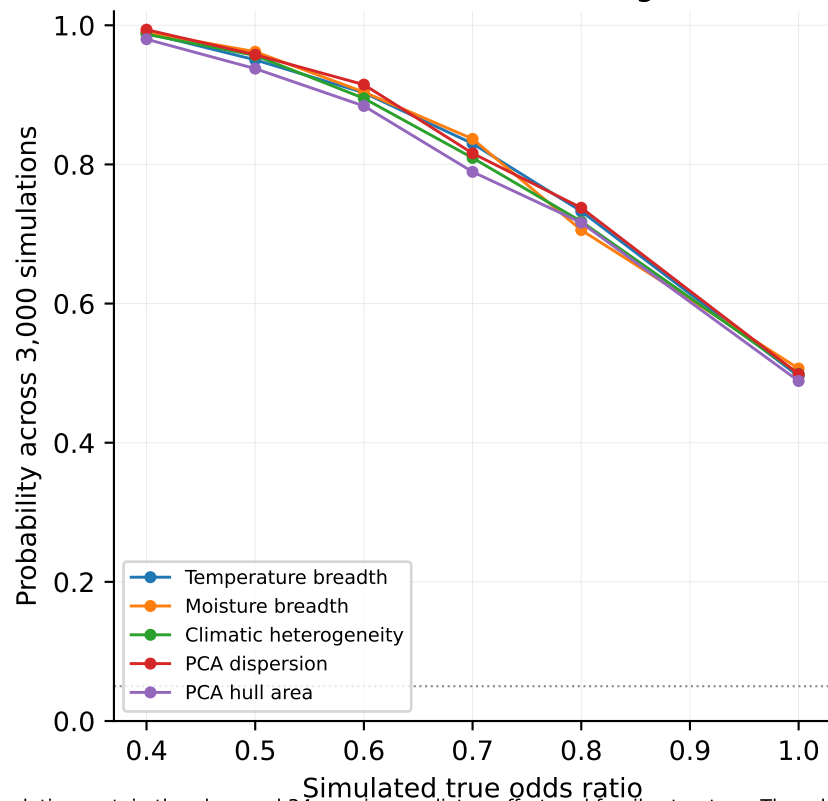
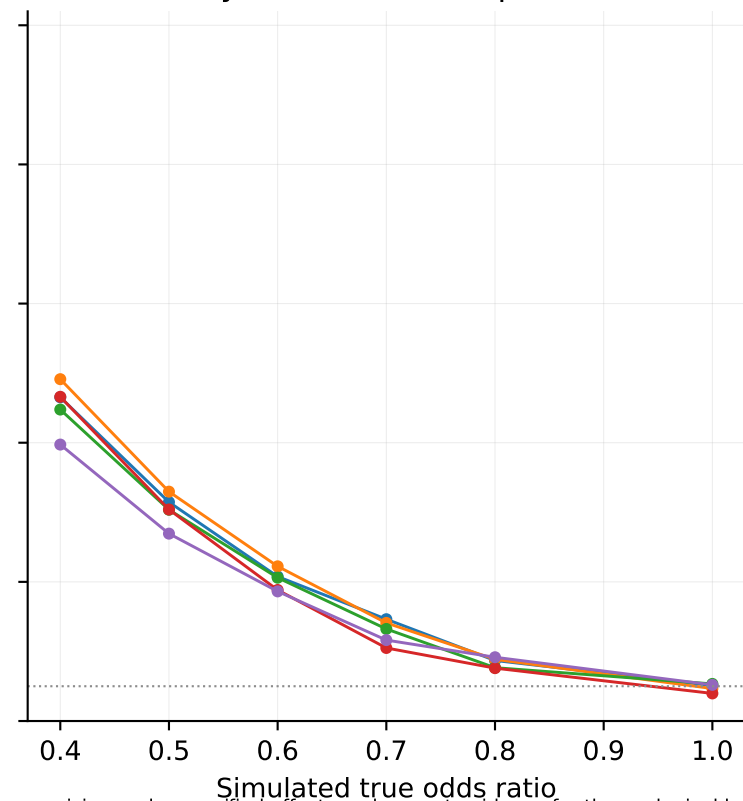


## Supporting Figure S2. Finite-sample design diagnostic

### Estimated coefficient is negative



### Family-clustered Wald $p < 0.05$



Simulations retain the observed 34-species predictor, effort and family structure. They describe design precision under specified effects and are not evidence for the ecological hypothesis.